$$\tau' = \beta\tau = \alpha^{3/2}\tau \tag{8}$$

$$\frac{\tau'^2}{l'^3} = \frac{\tau^2}{l^3}\ . \tag{9}$$

$\boldsymbol{r} = \boldsymbol{r}_{\mathrm{B}} - \boldsymbol{r}_{\mathrm{A}}$ $\boldsymbol{e}_r = \boldsymbol{r}/r$

$$\begin{cases} m\boldsymbol{a}_{\mathrm{A}} = -\dfrac{kq^2}{r^2}\left(\boldsymbol{e}_r \cdot \boldsymbol{i}\right)\boldsymbol{i} \\ m\boldsymbol{a}_{\mathrm{B}} = \dfrac{kq^2}{r^2}\left(\boldsymbol{e}_r \cdot \boldsymbol{j}\right)\boldsymbol{j} \end{cases}, \tag{1}$$

$$m\boldsymbol{a} = \frac{kq^2}{r^2}\boldsymbol{e}_r\ , \tag{2}$$

$\boldsymbol{a} = \boldsymbol{a}_{\mathrm{B}} - \boldsymbol{a}_{\mathrm{A}}$ B A (2) B A

m

$$U = \frac{kq^2}{r}\ . \tag{3}$$

B

A

$$L = \left|-md\boldsymbol{i} \times \left(-u_0\boldsymbol{j} + u_0\boldsymbol{i}\right)\right| = md\sqrt{\frac{kq^2}{md}}\ , \tag{4}$$

$$E = \frac{1}{2}m\left(-u_0\boldsymbol{j} + u_0\boldsymbol{i}\right)^2 + \frac{kq^2}{d} = \frac{2kq^2}{d}\ . \tag{5}$$

r_{m} u_{m} $r_{\mathrm{m}}\boldsymbol{e}_r$

$$mr_{\mathrm{m}}u_{\mathrm{m}} = md\sqrt{\frac{kq^2}{md}}\ , \tag{6}$$

$$\frac{1}{2}mu_{\mathrm{m}}^2 + \frac{kq^2}{r_{\mathrm{m}}} = \frac{2kq^2}{d}\ . \tag{7}$$